MAOMi

Faster simulation. Open optical interfaces. Ground truth from a constructed world.

1. Speed

188.7x	0.05219%	4225 → 1024
Measured acceleration	Final-image relative L2	Propagation locations

Original MATLAB CPU, 4225 sites: 85.303 s. MAOMi CUDA, 1024 sites: 0.452 s.

Same tissue/scattering input and MATLAB pupil; full-image relative L2 error 0.05219%, below 0.2%. Median of three subsequent calls on Ryzen 9 7950X / RTX 3080. Optical stage includes phase preparation, propagation, both tails and mask assembly; input/pupil generation, static scanning, I/O and startup excluded. Measured 17 October 2026.

2. Extensibility

MAOMi combines MATLAB-referenced tissue, activity, scattering and movie workflows with replaceable optical providers. Physical PSF data and image formation have separate interfaces: change the optics without rewriting the renderer. External-PSF workflows run on CPU; fresh native tissue propagation uses CUDA.

Area	MAOMi	NAOMi.jl
Execution	Batched CUDA propagation and fused kernels.	CPU / FFTW implementation.
Tissue optics	Scattering propagation and multiple excitation modes.	Cortical driver deferred; example uses a Gaussian PSF.

NAOMi.jl snapshot: 17 May 2026. Source comparison only; no Julia speed benchmark.

3. Connections and capabilities

Connection	Available now
RayOptics	Working adapter: lens and wavefront traces become custom PSFs or local PSF fields.
Wave optics / measured PSFs	Callable and NPZ inputs accept sampled intensity kernels with explicit units and sampling.
Procedural cells / Python	Editable 2-D cells produce clean/noisy pairs, latent cell maps, seeds and scene-disjoint splits.

The external PSF renderer currently forms 2-D intensity images; the native 3-D tissue route remains separate.

4. What's next

Extend to editable 3-D cells and dynamics, depth- and wavelength-dependent optics, more wave solvers and detectors. Fit morphology, fluorescence and instrument parameters to real references; generate paired ground truth for denoising and reconstruction, then validate on held-out experiments. The goal is a programmable ground-truth platform for instrument design and imaging models. Full 3-D customization, real-data calibration and model training are planned capabilities.